

Avinash Singh

Intern Scania/Traton in Autonomous Driving Research

+46 767027397 · avinjsingh@gmail.com · www.linkedin.com/in/avinashsinghsu · <https://avsngn-git.github.io/>

An analytically driven statistician with a deep foundation in mathematical statistics and machine learning from Stockholm University's Department of Mathematics. Specializes in analyzing model stability, investigating underlying statistical assumptions, and implementing uncertainty quantification frameworks. Proven experience in reasoning about complex systems and evaluating model limitations to deliver robust, interpretable data solutions.

SKILLS

Statistical Foundations: Probability Theory, Statistical Inference, Uncertainty Quantification, Conformal Prediction, Hypothesis Testing, Statistical Analysis

Machine Learning: Statistical Learning, Deep Learning, Reinforcement Learning, Unsupervised Learning, Dimensional Reduction, Transfer Learning, Predictive Modeling

Software & Ecosystem: Python, SQL, PyTorch, Pandas, NumPy, Apache Spark, Linux, Git, AWS

PROFESSIONAL EXPERIENCE

- **Traton / Scania Group** Södertälje, Sweden
Master's Thesis Worker and Summer Job – Autonomous Driving Post-Perception & Planning Research Team
01/2026 – Present
Investigated the feasibility of End-to-End self-driving models, trained for passenger cars adapting to heavy-duty vehicles. Added reliable uncertainty estimation to its trajectory predictions, showed that adapting an existing model was more stable and efficient than rebuilding one from scratch, and created realistic synthetic driving scenarios using gaussian splatting to test rare and safety-critical situations.
- **Alyst AB** Stockholm, Sweden
Summer Data Science Intern 06/2025 – 09/2025 Built an automated system that gathers information from email conversations and turns hundreds of messages into clear weekly summaries. Improved its accuracy and reliability so it could consistently handle incomplete information without losing important details.

METHODOLOGICAL & TECHNICAL PROJECTS

- **Beyond Vanilla: A Large-Scale Empirical Study of Transformer Architectures**
 - Built a PyTorch research platform comparing 10 Transformer architectures—including modern dense, grouped-query, sliding-window, linear-attention, and Mixture-of-Experts models—under controlled data, parameter, and compute budgets.
 - Orchestrated a five-seed, 50-run benchmark covering approximately 25 billion FineWeb-Edu tokens, with reproducible dataset preparation, verified checkpoints, automatic recovery, gradient monitoring, and experiment provenance.
 - Evaluated model quality, training and inference throughput, memory efficiency, long-context retrieval, and internal behavior using attention analysis, representation similarity, routing diagnostics, and statistical comparisons.
- **Comparative Analysis of Reinforcement Learning Policies for Game AI**

- Conducted a thorough empirical evaluation comparing Proximal Policy Optimization (PPO) and Deep Q-Networks (DQN) within complex, simulated environments.
 - Investigated the root algorithmic causes behind convergence behaviors, mapping PPO's 40% performance advantage directly to the mathematical clipping of its policy updates.
 - Validated model stability limits to demonstrate how constrained policy updates systematically prevent the chaotic divergence typical of standard deep Q-learning methods.
- **Semantic Segmentation of Aerial Drone Imagery**
 - Built a U-Net encoder-decoder network optimized for dense, pixel-level categorical classification of highly variable geographical terrain.
 - Evaluated model error distributions across high-contrast spatial boundaries (architectural, living, and environmental classes) to ensure categorical robustness.

EDUCATION

- **Stockholm University** Stockholm, Sweden
M.Sc. in Mathematical Statistics and Machine Learning – Department of Mathematics 09/2024 – Present
 - Rigorous emphasis on core probability theory, statistical inference, data structures, and the mathematical properties governing deep neural architectures.
- **National Institute of Technology (NIT)** Srinagar, India
B.Tech in Electrical Engineering 2011 – 2015
 - Completed core curriculum focused heavily on linear algebra, calculus, numerical methods, and differential modeling.